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Processos Estocásticos

(Dia 14/09/2026)

① Lem S finito. Semigrupo estocástico $\{P_t: t \geq 0\}$ padrão
 $\exists (X(t))_{t \geq 0}$ ^{Cadeia de Markov} ~~tal que~~ càdlàg tal que $\{P_t: t \geq 0\}$ é semigrupo
 de transição de $(X(t))_{t \geq 0}$.

prova: Seja G o gerador. $P_t = e^{tG}$. Se $i \neq j$,

$$g_{ij} = \lim_{h \downarrow 0} \frac{P_{ij}(h)}{h} \geq 0$$

$$\bullet \quad G\mathbb{1} = \lim_{h \downarrow 0} \frac{P_h \mathbb{1} - \mathbb{1}}{h} = 0$$

$$\Rightarrow q_i := -g_{ii} = \sum_{\substack{j \in S \\ j \neq i}} g_{ij} \geq 0$$

$$\bullet \quad \text{Escolha } \lambda > \max_{i \in S} q_i. \quad \text{Seja } R = I + \frac{1}{\lambda} G.$$

$$\Rightarrow R_{ij} \geq 0 \quad \forall i, j \quad \bullet \quad R\mathbb{1} = I\mathbb{1} + \frac{1}{\lambda} G\mathbb{1} = I\mathbb{1} = \mathbb{1}.$$

$\Rightarrow R$ estocástica.

$\bullet$ Seja $(Y_n)_{n \geq 0} \sim \text{Markov}(\lambda, R)$ (tempo discreto)

$\bullet$ Seja $(N(t))_{t \geq 0}$ processo de Poisson com taxa λ .

$\bullet$ Defina $X(t) := Y_{N(t)}, \quad \forall t \geq 0.$

Q2. $(X(t))_{t \geq 0}$ é Cadeia de Markov

$$\mathbb{P} \left(X(t_n) = j_n \mid \begin{array}{l} X(t_1) = j_1, \dots, X(t_{n-1}) = j_{n-1} \\ N(t_1) = m_1, \dots, N(t_{n-1}) = m_{n-1}, N(t_n) = m_n \end{array} \right)$$

$$= \mathbb{P} \left(Y_{m_n} = j_n \mid \begin{array}{l} Y_{m_1} = j_1, \dots, Y_{m_{n-1}} = j_{n-1} \\ N(t_1) = m_1, \dots, N(t_{n-1}) = m_{n-1}, N(t_n) = m_n \end{array} \right)$$

$$= \mathbb{P} \left(Y_{m_n} = j_n \mid Y_{m_{n-1}} = j_{n-1} \right)$$

• $(\tilde{P}_t)_{ij} =$
 $\cdot = \mathbb{P}(X(t) = j \mid X(0) = i) = \sum_{n \geq 0} \mathbb{P}(X(t) = j \mid X(0) = i, N(t) = n) \cdot \frac{(\lambda t)^n e^{-\lambda t}}{n!}$

• $\mathbb{P}(X(t) = j \mid X(0) = i, N(t) = n) = \mathbb{P}(Y_n = j \mid Y_0 = i)$

$$= \mathbb{P}(Y_n = j \mid Y_0 = i) = (R^n)_{ij}$$

$$\Rightarrow \tilde{P}_t = \sum_{n \geq 0} R^n \cdot \frac{(\lambda t)^n e^{-\lambda t}}{n!} = \cancel{R^n} \cdot e^{\lambda t R} e^{-\lambda t}$$

$$= e^{\lambda t(R-I)} = e^{tG} = P_t.$$



③ Lem (Kolmogorov's forward and backward equations)

S finito. $\{P_t: t \geq 0\}$ padrão semigrupo estocástico, gerador G .

Então, $\forall t \geq 0 \quad \frac{d}{dt} P_t = G P_t, P_0 = I$

$\frac{d}{dt} P_t = P_t G, P_0 = I.$

• π é distribuição estacionária se $\pi_i \geq 0, \langle \pi, \mathbf{1} \rangle = 1$.

Lem: $\{P_t: t \geq 0\}$ semigrupo estocástico padrão, gerador G . π estacionário

então ~~$\pi G = 0$~~ . $\pi G = 0$

Prova:

$0 = \frac{d}{dt} (\mathbf{1}) = \frac{d}{dt} (\pi P_t) = \pi \frac{d}{dt} P_t = \pi G P_t = \pi G$

Então existe

Lem: S finito, $(X(t))_{t \geq 0}$ càdlàg, cadeia de Markov. G gerador. Seja

$\tau_i = \inf\{t > 0 : X(t) \neq i\}$. $q_i := -g_{ii} = \sum_{\substack{j \in S \\ j \neq i}} g_{ij} \geq 0$

$P(\tau_i > t | X(0) = i) = e^{g_{ii}t} = e^{-q_i t}$

prova: càdlàg $\Rightarrow (P_t)_{t \geq 0}$ padrão.

• $q_i = -g_{ii} = \sum_{\substack{j \in S \\ j \neq i}} g_{ij} \geq 0$

• Fixado $t > 0$. ~~Dado~~ $n \geq 0$, $A_n := \bigcap_{k=0}^{2^n} \left\{ X\left(\frac{k t}{2^n}\right) = i \right\}$

$\Rightarrow A_{n+1} \subseteq A_n$

• Dado $X(0) = i$, $\{\tau_i > t\} = \bigcap_{n=0}^{\infty} A_n$ ($\Leftarrow$ càdlàg)

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$$\bullet P(A_n | X(0)=i) = P_{ii}\left(\frac{t}{2^n}\right)^{2^n}$$

$$\bullet \text{Gerador existe} \Rightarrow P_{ii}(h) = 1 + g_{ii}(h) + o(h) = 1 - q_i h + o(h) \text{ quando } h \rightarrow 0$$

$$\bullet P(\tau_i > t | X(0)=i) = \lim_{n \rightarrow \infty} P_{ii}\left(\frac{t}{2^n}\right)^{2^n}$$

$$= \lim_{n \rightarrow \infty} \left(1 + g_{ii}\left(\frac{t}{2^n}\right) + o\left(\frac{t}{2^n}\right)\right)^{2^n}$$

$$= e^{g_{ii}t} = e^{-q_i t}$$

$$\bullet \text{Se } q_i = 0 \dots \Rightarrow \tau_i = \infty \text{ condicional em } X(0)=i.$$

Exemplo (revisitado)

$$\bullet S = \{0, 1\}, \mu, \lambda > 0.$$

$$\bullet P(T_1, T_2 \leq h) = o(h) \quad \cancel{P(T_1 \leq h, T_2 \leq h)} \quad P(T_1 \leq h, T_2 > h) = \lambda h + o(h)$$

$$\Rightarrow P_h = \begin{pmatrix} 1 - \lambda h + o(h) & \lambda h + o(h) \\ \mu h + o(h) & 1 - \mu h + o(h) \end{pmatrix}$$

$$\Rightarrow G = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix} \quad \text{Obs: } G^2 = G \cdot (-\lambda - \mu).$$

$$P_t = e^{tG} = I + \sum_{n=1}^{\infty} \frac{t^n}{n!} (-\lambda - \mu)^{n-1} G = I + \frac{1 - e^{-(\lambda + \mu)t}}{\lambda + \mu} G$$

⑤ Processo de Poisson

- $\lambda > 0$
- $\tau_1, \tau_2, \dots$ iid $\text{Exp}(\lambda)$
- $T_0 = 0, T_k = \sum_{j=1}^k \tau_j$
- ~~XXXX~~ $X(t) = \sum_{k=1}^{\infty} \mathbb{1}_{\{T_k \leq t\}}$
- $(X(t))_{t \geq 0}$: processo de poisson com taxa λ .
- $\Rightarrow (X(t))_{t \geq 0}$ é cadeia de Markov

~~XXXX~~ $X(t) = \min\{m, X(s)\}$

Teo:

- (i) $X(0) = 0$
- (ii) $X(t) - X(s)$ é independente de $X(s)$ $\forall s < t$.
- (iii) $P(X(t) = 0) = 1 - \lambda t + o(t)$
- (iv) $P(X(t) = 1) = \lambda t + o(t)$
- (v) $P(X(t) \geq 2) = o(t)$

prova: (i) ✓

$$\begin{aligned}
 \text{(ii)} \quad & P(T_{n+1} - s > t, X(s) = n) = P(T_n \leq s, T_{n+1} > s+t) \\
 & = \mathbb{E}(\mathbb{1}_{\{T_n \leq s\}} \mathbb{E}(\mathbb{1}_{\{T_{n+1} > s+t\}} | T_n)) \\
 & = \mathbb{E}(\mathbb{1}_{\{T_n \leq s\}} e^{-\lambda(s+t-T_n)}) \\
 & = e^{-\lambda t} \mathbb{E}(\mathbb{1}_{\{T_n \leq s\}} e^{-\lambda(s-T_n)})
 \end{aligned}$$

$$\textcircled{6} = e^{-\lambda t} P(T_n \leq s, T_{n+1} > s) = e^{-\lambda t} \cdot P(X(s) = n)$$

$$\Rightarrow P(T_{n+1} - s > t \mid X(s) = n) = e^{-\lambda t}$$

Dado ~~Given~~ $X(s) = n$, $\begin{pmatrix} T_{n+1} - s \\ T_{n+2} \\ T_{n+3} \\ \vdots \end{pmatrix}$ tem mesma distribuição

que $\begin{pmatrix} T_1 \\ T_2 \\ T_3 \\ \vdots \end{pmatrix}$.

$$\text{Dado } X(s) = n, \quad X(t) - X(s) = \mathbb{1}\{T_n - s \leq t\} + \sum_{j \geq 2} \mathbb{1}\{T_{n+j} \leq t\}$$

$$X(t) - X(s) = \sum_{j \geq n+1} \mathbb{1}\{T_{k+j} \leq t\}$$

$$= \sum_{j \geq n+1} \mathbb{1}\{T_{k+j} - s \leq t - s\}$$

$$\text{Dado } X(s) = n, \quad (X(t) - X(s))_{t \geq s} \mid X(s) = n \sim (X(t))_{t \geq 0}$$

(i) ■

⑦

$$(iii) P(X(t)=0) = P(\tau_1 > t) = e^{-\lambda t} = 1 - \lambda t + o(t).$$

~~$$P(X(t)=1) = P(\tau_1 \leq t, \tau_1 + \tau_2 > t)$$~~

~~$$P(X(t)=1) = P(\tau_1 \leq t) = 1 - e^{-\lambda t}$$~~

~~$$(v) P(X(t) \geq 2) = P(\tau_1 + \tau_2 \leq t) = 1 - P(\tau_1 + \tau_2 > t)$$~~

~~$$= 1 - P(\tau_1 > t/2, \tau_2 > t/2) = 1 - e^{-\lambda t}$$~~

$$\leq P(\tau_1 \leq t, \tau_2 \leq t) = (1 - e^{-\lambda t})^2 = o(t)$$

$$(iv) P(X(t)=1) = 1 - P(X(t)=0 \text{ ou } X(t) \geq 2)$$

$$= 1 - (1 - \lambda t + o(t)) - o(t) = \lambda \cdot t + o(t)$$

Seja $X^{(m)}(t) = X(t) \wedge m = \min\{X(t), m\}$

$$\Rightarrow (X^{(m)}(t))_{t \geq 0} \text{ é cadeia de Markov em } \{0, 1, \dots, m\}.$$

$$\Rightarrow P(X^{(m)}(t)=j | X^{(m)}(s)=i) = P(X(t)=j | X(s)=i)$$

$$\forall s < t, \quad \forall i, j < m.$$

Seja $(P_t^{(m)})_{t \geq 0}$ o semigrupo de transição de $(X_t^{(m)})_{t \geq 0}$.

$$\Rightarrow (P_t^{(m)})_{t \geq 0} \text{ é padrão} \Rightarrow \text{tem gerador } G^{(m)}, P_t^{(m)} = e^{tG^{(m)}}.$$

⑧ Último teorema implica

$$G^{(m)} = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & \dots & 0 \\ 0 & -\lambda & \lambda & 0 & \dots & 0 \\ 0 & 0 & -\lambda & \lambda & \dots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \ddots & -\lambda \lambda \\ 0 & 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

note que

note que

$$G_{m,m}^{(r_m)} = \lim_{h \downarrow 0} \frac{\mathbb{P}(X(h) = m | X(0) = m) - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1-1}{h} = 0.$$

Seja $J_0^{(n)} = \begin{pmatrix} 1 & & & 0 \\ & 1 & & \\ & & \ddots & \\ 0 & & & 1 & \\ & & & & 0 \end{pmatrix}$, $J_K^{(n)} = \begin{pmatrix} 0 & \dots & 0 & 1 & 0 & 0 \\ & & & 0 & 1 & 0 \\ & & & & \ddots & 1 \\ & & & & & 0 \\ & & & & & 0 \end{pmatrix}$ (K vezes)

$$\Rightarrow J_i^{(m)} \cdot J_K^{(m)} = J_{i+K}^{(m)} \quad \forall i, K \geq 0$$

$$J_k^{(m)} = 0 \quad \forall k \geq m+1.$$

$$G^{(m)} = -\lambda J_0 + \lambda J_1 = -(\lambda)(J_0 - J_1)$$

$\forall j < m,$

$$\forall j < m, \quad P(X(t) = j) = P(X^{(m)}(t) = j) \quad \text{[crossed out]} \quad \text{[crossed out]}$$

~~$$= \sum_{k=0}^{\infty} \left(\sum_{l=0}^{\infty} \binom{k}{l} (-1)^l \right) \sum_{l=0}^{\infty} \binom{k}{l} (-1)^l \frac{1}{2^k}$$~~

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Capítulo 2 (das notas do VS)

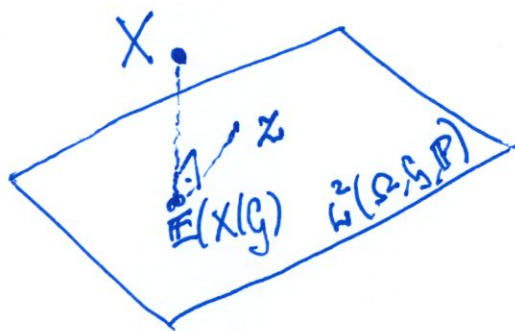
Esperança Condicional

- $E(X) = \operatorname{argmin}_{x \in \mathbb{R}} E((X-x)^2)$, ~~$E(X) = \int X dP$~~ .
se $X \in L^2(\Omega, \mathcal{F}, P)$.

Def: Seja $\mathcal{G} \subseteq \mathcal{F}$, $X \in L^2(\Omega, \mathcal{F}, P)$.

define $E(X|\mathcal{G}) = \operatorname{argmin}_{Z \in L^2(\Omega, \mathcal{G}, P)} E((X-Z)^2)$

isto é $E(X|\mathcal{G}) = \text{Orthogonal Proj. of } X \text{ on } L^2(\Omega, \mathcal{G}, P)$.



$$E((X - E(X|\mathcal{G})) \cdot Z) = 0, \quad \forall Z \in L^2(\Omega, \mathcal{G}, P).$$